SYNOPSIS on

TITLE – CUSTOMER SHOPPING DATA ANALYSIS

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ABOUT THE TOPIC (DATASET)

This project analyzes the Customer Shopping Dataset, which records approximately 99,000 retail transactions across 10 shopping malls in Istanbul between 2021 and 2023. Each row represents a single line-item purchase, capturing customer demographics, product category, payment method, and transaction value – enabling analysis of customer behavior, sales trends, and mall-level performance.

Data Structure

The cleaned dataset (customer_shopping_cleaned.csv) is organized in a tabular format, where each row represents a customer transaction on a specific date. It contains the original 10 columns plus 4 columns engineered during preprocessing (total_amount, year, month, day_of_week). The columns are:

Column Name	What Information It Carries
invoice_no	Unique identifier for each transaction or invoice. Each value represents a distinct sale.
customer_id	Unique identifier for customers. Each value represents a different customer who has made a purchase.
gender	The gender of the customer. Contains categorical values such as 'Male' or 'Female'.
age	The age of the customer, in years. Missing values were imputed with the column median during cleaning.
category	The category of the item purchased (e.g. Clothing, Cosmetics, Food & Beverage, Technology).
quantity	The number of units of the item purchased in the transaction.
price	The price of a single unit of the item purchased.
payment_method	The method used to make the payment: 'Cash', 'Credit Card', or 'Debit Card'.
invoice_date	The date the transaction occurred.
shopping_mall	The shopping mall where the transaction took place.
total_amount	Derived during cleaning as quantity × price – the total value of the transaction.
year	Derived from invoice_date – the calendar year the transaction occurred, for year-over-year comparison.

Column Name	What Information It Carries
month	Derived from invoice_date – the calendar month (1–12) the transaction occurred, for seasonal analysis.
day_of_week	Derived from invoice_date – the name of the weekday (e.g. Monday) the transaction occurred, for weekday-pattern analysis.

These columns collectively provide detailed information about each transaction, including who made the purchase, what was purchased, how much was purchased, the cost, how it was paid for, and where and when the transaction occurred. This dataset is useful for analyzing customer behavior, sales trends, inventory management, and financial performance of the shopping malls.

Data Preprocessing

1. **Data Cleaning:** Removed duplicate transaction records and imputed missing age values using the column median, to avoid distorting the age distribution.
2. **Feature Engineering:** Derived a total_amount field (quantity × price) and extracted year, month, and day_of_week from invoice_date to support seasonal analysis.
3. **Normalization:** Applied Min-Max scaling (scikit-learn's MinMaxScaler) to age, quantity, price, and total_amount so numeric features are on a comparable scale.
4. **Encoding Categorical Variables:** Applied one-hot encoding to gender, payment_method, and category, producing a model-ready dataset alongside the human-readable cleaned dataset.

Implementation Steps

1. **Data Ingestion:** Load the dataset into a Python data analysis environment.
2. **Preprocessing:** Clean and prepare the data for analysis (see Data Preprocessing above).
3. **Exploratory Data Analysis (EDA):** Conduct EDA to understand data distribution and initial patterns.
4. **Modeling (Future Scope):** The dataset is preprocessed (normalized and one-hot encoded) and structured to support future classification, clustering, or time-series modeling with scikit-learn; the current implementation focuses on cleaning, EDA, and reporting rather than predictive modeling.
5. **Visualization:** Produce EDA charts covering sales by category, age/gender distribution, payment method share, monthly sales trend, mall-level performance comparison, and a correlation heatmap of numeric features (Matplotlib/Seaborn).
6. **Reporting:** Load the cleaned data into a MySQL database via mysql-connector-python, then connect Power BI to build an interactive dashboard with KPI cards, category/mall comparisons, and time-based slicers for stakeholder reporting.

Data set: customer_shopping_data

Technologies: pandas, numpy, matplotlib, seaborn, scikit-learn, mysql-connector-python, Microsoft Excel, Microsoft Power BI.

Software Requirements

Operating System – Windows, Linux and macOS

IDE – Jupyter Notebook

Hardware Requirements

RAM – 8 GB minimum, 16 GB or more recommended
Processor – Intel Core i5 or equivalent minimum, Intel Core i7 or higher recommended